BRAC UNIVERSITY

Quiz 1 **Department of Computer Science and Engineering**

CSE110: Programming Language I Duration: 40 minutes Marks: 20

Name: (Please write in CAPITAL LETTERS)	ID:	Section: 42

1. Write a python program that will take an integer number of at least two digits from the user and check if the number is a magic number or not. An integer is said to be a magic number if the last digit of the number is **either divisible by 3 or 2 but not both.**
   [Consider the input number to be an INTEGER. You are not allowed to use String indexing for solving this task] [Marks 10]
   [CO 3 + CO 4]

Sample Input 1:
324
Sample output 1:
The last digit is 4
324 is a Magic number
Explanation:
324 is a magic number because the last digit of the number is 4 which is divisible by 2.

Sample Input 2:
7446
Sample output 2:
The last digit is 6
7446 is not a Magic number
Explanation:
7446 is not a magic number because the last digit of the number is 6 which is divisible by both 3 and 2.

Sample Input 3:
25
Sample output 3:
The last digit is 5
25 is not a Magic number
Explanation:
25 is not a magic number because the last digit of the number is 5 which is not divisible by either 3 or 2.

1. Trace the below table and write the outputs in the given boxes. You MUST show the trace table. [Marks 10] [C0 3 + CO 4]

1	`p, sum , z = 0 , 20, 70`
2	`while (z > 7):`
3	`  while ( sum < z+1 ):`
4	`    p = z + sum -29`
5	`    print(3 + int (p / 5))`
6	`    sum += 10`
7	`  z -= 15`
8	`print(False and (True or not True) or not False)`
9	`print(sum // 7 + 2)`
10	`print(z+2 > 9 == False)`
11	`print(int(69/7 + 3))`

Output